

Two epsilons and the conditioning of G

recap of the September 2026 changes

30 September 2026

The phrase “*I removed epsilon*” mixed up two different constants. This note separates them, lists what changed in each, and summarises what we know about the conditioning of the Gram matrix.

1 Setup and notation

At step k , MGD solves two linear systems on the walker ensemble x_k :

$$G_k \eta_k = \dot{\phi}(I_{t_k}), \quad G_k \theta_k = r_k, \quad (G_k)_{nm} = \mathbb{E}_{x_k}[\langle \nabla \phi_n(x), \nabla \phi_m(x) \rangle].$$

G_k is the **Gram matrix of the gradients** $\nabla \phi_n$ in $L^2(p_k)$. So

$$G_k \text{ singular} \iff \exists c \neq 0 : \sum_n c_n \nabla \phi_n = 0 \text{ } p_k\text{-a.s.} \iff \sum_n c_n \phi_n \text{ is constant on the support.}$$

The ϕ_n are nonlinear in x , but this is linear dependence between the *functions* $\nabla \phi_n$. Below I call it “(near) linear dependence of the gradients” rather than “collinearity of the potentials”.

2 Epsilon #1: the Tikhonov ridge on G (unchanged)

Every per-step solve (`compute_eta`, `compute_theta`) uses Jacobi scaling plus a ridge:

$$D = \text{diag}(G_k), \quad \tilde{G}_k = D^{-1/2} G_k D^{-1/2} + \lambda I, \quad \eta_k = D^{-1/2} \tilde{G}_k^{-1} D^{-1/2} \dot{\phi}.$$

Here λ is `--regularization`. The current runs use $\lambda = 10^{-4}$ (labels `reg1e-4`) or 10^{-2} . **It has never been removed.**

It is already a “vector epsilon”. In the original coordinates,

$$\tilde{G}_k = D^{-1/2} (G_k + \lambda D) D^{-1/2},$$

so the ridge on coordinate n is $\lambda \mathbb{E}_{x_k} |\nabla \phi_n|^2$: it scales with the statistic. Also, `stat_scale` normalises each scalar statistic at fit time so that

$$\mathbb{E}_{\text{data}} |\nabla \phi_n|^2 = 1,$$

which is the proposed normalisation $\mathbb{E} |\nabla \phi_n| = 1$, taken in L^2 instead of L^1 .

3 Epsilon #2: the $|z|$ floor inside the scalar potentials (replaced)

`Scalar_GGD_KRegion` splits each wavelet channel $z_j = \psi_j * x$ into regions of $|z_j|$ separated by cuts $c_{j,1} < c_{j,2} < \dots$, fitted on the *raw* $|z_j|$. Forward and gradient use a smoothed modulus:

$$|z|_\varepsilon = \sqrt{z^2 + \varepsilon},$$

so that the core gradient $\alpha z |z|^{\alpha-2}$ stays finite at 0 when $\alpha < 1$.

The bug. With $\varepsilon = 10^{-6}$ we get $|z|_\varepsilon \geq 10^{-3}$. The finest channels live at $|z| \sim 10^{-4}$: for ψ channels 0–7, *all* cuts are below 10^{-3} . So every point landed above the cuts.

channel 0, region	1	2	3	4
share in the data	40%	50%	8%	2%
share seen by ϕ	0%	0%	0.5%	99.5%

The inner regions had $\nabla \phi_n \approx 0$ or identical gradients. G was therefore rank-deficient *on the data*, and `prune_collinear` dropped these statistics: 52 of 81 in the full fit, keeping 1 of 77 on a 1000-series fit. The near-zero mass (sparsity) and the fine-scale tails were left unconstrained.

The fix (commit 27bd5b8): a per-channel floor

$$\varepsilon_j = (q_{10^{-3}}(|z_j|))^2,$$

computed on the data at fit time. It moves at most 0.1% of the points of each channel. Region shares seen by ϕ now match the fitted ones to within 1.4 points. The floor was replaced, not removed. Old fits reload with the scalar ε and reproduce.

The only genuine redundancy inside the ψ bank is ψ channel 21 vs 20 (filter overlap 1.0000, but not bit-identical copies). `--dedup_tol 1e-8` removes it. (Redundancies *between* the $Q = 1$ and $Q = 3$ banks were not checked here; see the update of 1 October in the conditioning section.)

4 What changed around the ridge: masking near-empty statistics

The per-channel floor makes regions that are empty on the walkers *nearly* dead rather than *exactly* dead. Early in the transport, walkers look like noise, and the outer regions of some channels get $G_{nn}(x_k) \sim 10^{-16}$ – 10^{-8} times their data value. For such a nearly decoupled coordinate, the scaled solve gives

$$\eta_n \approx \frac{\dot{\phi}_n}{G_{nn}(1 + \lambda)}.$$

The ridge doesn't help here: after Jacobi scaling it is relative, so a vanishing G_{nn} still produces a huge coefficient. Observed: drift up to 45–129 (vs 1.6–3.5 with the old statistics), and the walkers blew up at step 163 (job 268295).

Fix (commit 1c343be): at each step, solve only on the live set

$$L_k = \{ n : (G_k)_{nn} > \tau \cdot (G_{\text{data}})_{nn} \}, \quad \tau = 10^{-6},$$

and set $\eta_{k,n} = \theta_{k,n} = 0$ for $n \notin L_k$. Such a statistic cannot move the walkers at that step anyway. $\tau = 0$ for non-region potentials, which gives back the older exact-zero test (added for singular solves near $t = 1$, jobs 115127/115128). With the mask, $\tau = 10^{-6}$ removes the tail of large drifts and masks nothing from $t \approx 0.3$ on.

In Θ_{reg} (commit 62d2c0e): the same live mask is passed to the block-Thomas solve (dead potentials pinned to $\Theta_{k,n} = 0$). The data block uses the same ridge as the per-step solve, $M_k + \lambda \text{diag}(M_k)$. A mismatch there (offline $\lambda = 10^{-6}$ vs 10^{-4} in the run) was one cause of the Θ_{reg} explosions.

5 Conditioning of G : what we know

We use $\kappa(\tilde{G}) = \lambda_{\max}(\tilde{G})/\lambda_{\min}(\tilde{G})$ on the Jacobi-scaled matrix (the only one that is solved).

Scalar_psi_gaussianK only, data Gram, 1000 series	statistics kept	κ
old floor $\varepsilon = 10^{-6}$	1 / 77	—
per-channel floor	77 / 77	9.6×10^4
per-channel floor + dedup_tol 1e-8	73 / 73	11.9

These numbers are for the `Scalar_psi_gaussianK` statistics *alone* (73–77 of the $r = 281$ statistics). Once the floor is fixed and the near-copy filter removed, *this block* of the data Gram is well conditioned, and the large κ of this block came from one duplicated filter. The full 281×281 Gram, which also contains the other families and the near-dependencies *between* families, was not computed here; see the update below. The Jean Zay fit keeps 78/78 ψ statistics ($r = 281$ in total).

What we don’t know yet: $\kappa(\tilde{G}_k)$ **along the transport.** We haven’t measured it step by step. We know two regimes where G_k degenerates on the walkers:

- early t : near-empty regions ($G_{nn} \rightarrow 0$), now handled by the mask;
- $t \rightarrow 1$: exactly dead regions, also masked.

Next step: $\log \kappa(\tilde{G}_k|_{L_k})$ and $\lambda_{\min}$ at each step (or every N steps) in `compute_eta`, together with $|L_k|$. That tells us whether ill-conditioning in the bulk of the transport contributes to the numerical problems, and if so, whether to use a per-coordinate λ , a truncated or early-stopped solve, or further masking. This needs a new Jean Zay run.

Update, 1 October 2026. The run with this logging (job 382722, 30 September, every 10 steps) has been analysed; details in `notes/gram_instability_1001/gram_instability.pdf`. In short:

- The full Jacobi-scaled $\tilde{G}_k$ (all 281 statistics) has three directions far weaker than all the others, at almost every step from $t = 0$ to $t = 1$. The weakest is the near-copy pair $L_6[7] / L_6^\psi[20]$: the $Q = 1$ and $Q = 3$ banks put almost the same filter at the coarsest scale, and the existing deduplication only compares filters inside the $Q = 3$ bank.

- The two weakest eigenvalues ($\sim 10^{-7}$ – 10^{-6} , i.e. $\kappa \sim 10^7$ – 10^8) are at the precision limit of G , which is accumulated in `float32`; their exact values cannot be trusted. The fourth smallest is ~ 3 – $5 \cdot 10^{-4}$, so without the three weak directions κ would be at best ~ 2 – $5 \cdot 10^4$, roughly constant in t .
- The mask acted on 5 statistics, all before $t = 0.15$.

Take-home.

1. The Tikhonov ridge was never removed. The $|z|$ floor in the potential was made per-channel.
2. The old floor made fine-scale statistics degenerate. That, not a missing ridge, is what got them pruned.
3. A relative ridge cannot control statistics that become nearly empty on the walkers. Masking can.
4. The `Scalar_psi_gaussianK` block of the data Gram is now well conditioned ($\kappa \approx 12$). The full Gram along the transport is not (update of 1 October): three weak directions, at least the weakest due to near-copy filters across the $Q = 1$ and $Q = 3$ banks.